From a 2σ Peak to a Conservative LOSVD Separation

Variance, two-component mixtures, and CRD target selection

A detailed conceptual and mathematical derivation for MaNGA-to-KCWI/KCRM feasibility work

Main result. If the stellar spectrum at one position contains two components with light fractions f_A and $1 - f_A$, mean velocities V_A and V_B, and intrinsic dispersions σ_A and σ_B, then the variance of the combined LOSVD is

$$\boxed{\sigma_{\text{mix}}^2 = f_A\sigma_A^2 + (1 - f_A)\sigma_B^2 + f_A(1 - f_A)(V_A - V_B)^2.}$$

The first two terms describe the velocity spread *inside* the two disks. The final term describes extra broadening because the two disks are centered at different velocities. In a counter-rotating galaxy, this last term is the key source of the off-center 2σ enhancement.

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1 Why variance is the right starting point

The word *variance* can sound abstract, but the idea is simple: it measures how spread out a set of values is around its mean. For stellar kinematics, the values of interest are line-of-sight velocities.

Suppose, only as a toy example, that three equal-light groups of stars have velocities

$$70, \qquad 80, \qquad 90\ \mathrm{km\,s^{-1}}.$$

Their mean velocity is

$$\bar{V} = \frac{70+80+90}{3} = 80\ \mathrm{km\,s^{-1}}.$$

The deviations from the mean are

$$-10, \qquad 0, \qquad +10\ \mathrm{km\,s^{-1}}.$$

If we simply averaged the deviations, they would always cancel:

$$\frac{-10+0+10}{3} = 0.$$

That cannot measure spread. Variance solves this by squaring each distance from the mean before averaging:

$$\mathrm{Var}(V) = \frac{(-10)^2+0^2+(+10)^2}{3} = 66.7\ (\mathrm{km\,s^{-1}})^2.$$

The square root of the variance is the standard deviation,

$$\sigma = \sqrt{66.7} = 8.2\ \mathrm{km\,s^{-1}}.$$

Thus, in ordinary language:

> **Variance is the average squared distance of the velocities from their overall mean.** The velocity dispersion σ is the square root of that quantity, so it has the familiar units of km/s.

This is exactly the statistical quantity encoded by a Gaussian LOSVD width. We do not literally resolve individual stars in a MaNGA spaxel; instead, the integrated spectrum contains luminosity-weighted information from many stars. Conceptually, however, the LOSVD is a distribution of stellar velocities, and its variance describes the width of that distribution.

1.1 Why "distance from the overall mean" matters

Consider one stellar disk whose LOSVD is centered at 80 $\mathrm{km\,s^{-1}}$ with dispersion 20 $\mathrm{km\,s^{-1}}$. Relative to its own center, that disk has variance 20^2.

Now imagine placing that same disk inside a combined spectrum whose *overall* mean is 0 $\mathrm{km\,s^{-1}}$. Even if the disk itself is dynamically cold, all of its stars are also displaced by roughly 80 $\mathrm{km\,s^{-1}}$ from the combined mean. The combined distribution is therefore broad for two reasons:

1. stars within the disk are spread around the disk's own mean;
2. the disk's mean itself is displaced from the mean of the complete mixture.

That distinction is the heart of the two-disk derivation.

2 Set up the two-disk LOSVD

At one spatial location, let the spectrum contain light from two stellar disks.

	Disk A	Disk B
Light fraction	f_A	$1 - f_A$
Mean velocity	V_A	V_B
Intrinsic dispersion	σ_A	σ_B

The underlying velocity separation is

$$\boxed{\Delta V \equiv |V_A - V_B|.}$$

Figure 1 illustrates the idea. Each component has its own intrinsic width, but the component centers are also separated. A one-component model has to absorb both effects into a single broad LOSVD.

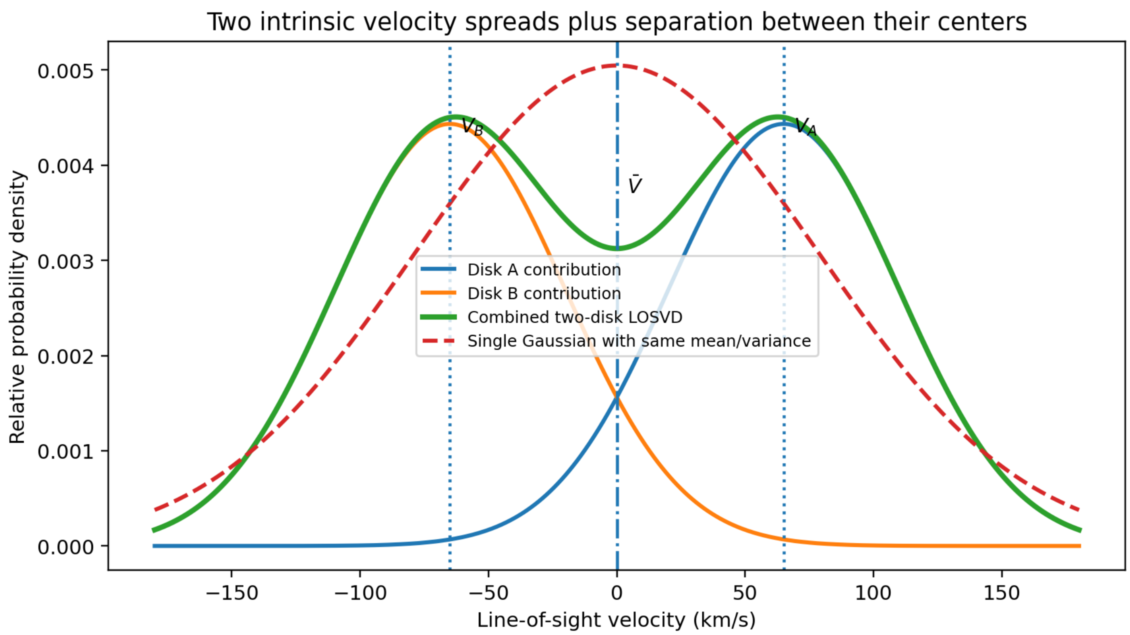

Figure 1: Conceptual two-component LOSVD. The widths of the individual curves represent intrinsic dispersions. Their different centers add an additional source of variance to the combined LOSVD. The dashed Gaussian has the same first two moments as the mixture; the true mixture need not itself be Gaussian.

3 Step 1: the mean velocity of the combined spectrum

The overall mean is just the luminosity-weighted average of the component means:

$$\boxed{\bar{V} = f_A V_A + (1 - f_A) V_B.}$$

This immediately explains why a single-component stellar velocity map can strongly suppress counter-rotation. If $f_A = 0.5$ and the two disks are symmetric about systemic, for example

$$V_A = +60, \qquad V_B = -60 \text{ km s}^{-1},$$

then

$$\bar{V} = 0.5(+60) + 0.5(-60) = 0.$$

Nothing about the underlying disks is moving slowly; their opposite contributions simply cancel in the first moment.

If the light fractions are 65/35 instead,

$$\bar{V} = 0.65(+60) + 0.35(-60) = +18 \text{ km s}^{-1}.$$

Thus an apparent single-component velocity of only $\sim 20 \text{ km s}^{-1}$ can coexist with two components separated by 120 km s^{-1}.

4 Step 2: write the variance of Disk A relative to the combined mean

Take an individual velocity belonging to Disk A. Write it as

$$v = V_A + \delta v_A,$$

where δv_A is the random velocity relative to Disk A's own mean. By definition,

$$\langle \delta v_A \rangle = 0, \qquad \langle \delta v_A^2 \rangle = \sigma_A^2.$$

Its displacement from the *overall* mean is

$$v - \bar{V} = (V_A - \bar{V}) + \delta v_A.$$

Square this:

$$\begin{aligned}(v - \bar{V})^2 &= \Big[(V_A - \bar{V}) + \delta v_A\Big]^2 \\ &= (V_A - \bar{V})^2 + 2(V_A - \bar{V})\delta v_A + \delta v_A^2.\end{aligned}$$

Now average over the Disk-A stars. The middle term vanishes because $\langle \delta v_A \rangle = 0$:

$$\begin{aligned}\Big\langle (v - \bar{V})^2 \Big\rangle_A &= (V_A - \bar{V})^2 + 2(V_A - \bar{V})\underbrace{\langle \delta v_A \rangle}_{0} + \underbrace{\langle \delta v_A^2 \rangle}_{\sigma_A^2} \\ &= \boxed{\sigma_A^2 + (V_A - \bar{V})^2.}\end{aligned}$$

This is a very useful conceptual equation. Disk A contributes width from two places:

$$\underbrace{\sigma_A^2}_{\text{spread inside Disk A}} \qquad + \qquad \underbrace{(V_A - \bar{V})^2}_{\text{Disk A's center is offset from the mixture mean}}.$$

Exactly the same argument for Disk B gives

$$\boxed{\Big\langle (v - \bar{V})^2 \Big\rangle_B = \sigma_B^2 + (V_B - \bar{V})^2.}$$

5 Step 3: luminosity-weight the two contributions

Disk A supplies fraction f_A of the light and Disk B supplies $1 - f_A$. Therefore the total variance is

$$\sigma^2_{\mathrm{mix}} = f_A \left[\sigma^2_A + (V_A - \bar{V})^2\right] + (1 - f_A) \left[\sigma^2_B + (V_B - \bar{V})^2\right]. \tag{1}$$

At this stage the physics is already visible. Equation 1 says:

> The total variance is the light-weighted intrinsic variance of each disk, plus the light-weighted squared displacement of each disk's center from the overall mean.

6 Step 4: simplify the offsets from the combined mean

Start with Disk A:

$$\begin{aligned} V_A - \bar{V} &= V_A - [f_A V_A + (1 - f_A) V_B] \\ &= (1 - f_A)(V_A - V_B). \end{aligned}$$

Similarly,

$$\begin{aligned} V_B - \bar{V} &= V_B - [f_A V_A + (1 - f_A) V_B] \\ &= -f_A (V_A - V_B). \end{aligned}$$

Substitute these into Equation 1:

$$\begin{aligned} \sigma^2_{\mathrm{mix}} = f_A \sigma^2_A &+ (1 - f_A)\sigma^2_B \\ &+ f_A (1 - f_A)^2 (V_A - V_B)^2 \\ &+ (1 - f_A) f_A^2 (V_A - V_B)^2. \end{aligned}$$

Factor the last two terms:

$$\begin{aligned} \sigma^2_{\mathrm{mix}} = f_A \sigma^2_A &+ (1 - f_A)\sigma^2_B \\ &+ f_A (1 - f_A)(V_A - V_B)^2 \left[(1 - f_A) + f_A\right]. \end{aligned}$$

Since $(1 - f_A) + f_A = 1$,

$$\boxed{\sigma^2_{\mathrm{mix}} = f_A \sigma^2_A + (1 - f_A)\sigma^2_B + f_A (1 - f_A)(V_A - V_B)^2.}$$

Equivalently, using $\Delta V = |V_A - V_B|$,

$$\boxed{\sigma^2_{\mathrm{mix}} = f_A \sigma^2_A + (1 - f_A)\sigma^2_B + f_A (1 - f_A)\Delta V^2.}$$

6.1 A shorter statistical interpretation: the law of total variance

The same result can be summarized in one sentence:

$$\boxed{\text{total variance} = \text{average variance within the components} + \text{variance of the component means.}}$$

For our two disks,

$$\underbrace{f_A \sigma^2_A + (1 - f_A)\sigma^2_B}_{\text{within-disk variance}} + \underbrace{f_A (1 - f_A)\Delta V^2}_{\text{between-disk variance}}.$$

This identity is known generally as the *law of total variance*. The detailed derivation above is simply that law written explicitly for two LOSVD components.

7 Why a 2σ peak is strongest near comparable light contributions

The counter-rotation term is

$$f_A(1 - f_A)\Delta V^2.$$

For a fixed ΔV, its dependence on light fraction is entirely contained in $f_A(1 - f_A)$. As Figure 2 shows, this factor is zero when either component supplies all the light and reaches its maximum value, 0.25, at $f_A = 0.5$.

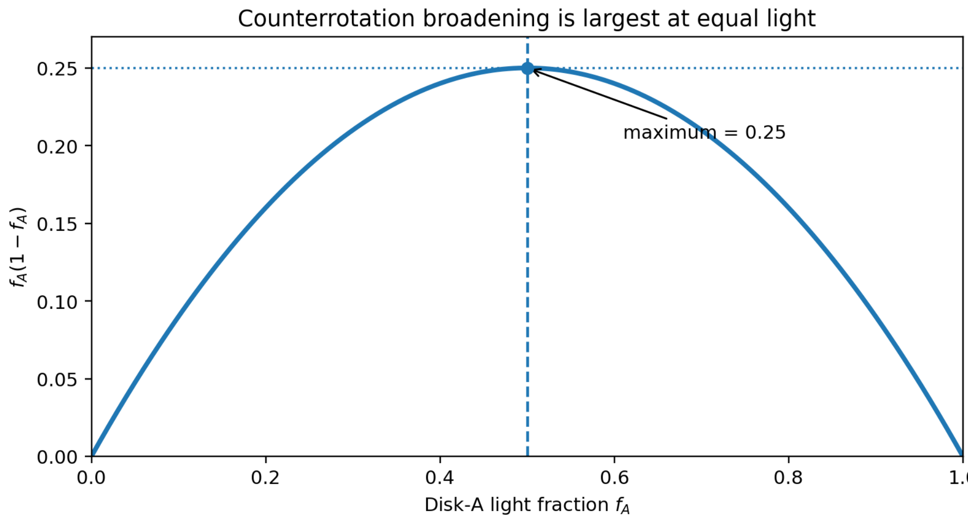

Figure 2: The factor $f_A(1 - f_A)$ is maximized at equal light. This is why the velocity-separation term contributes most strongly where both disks are important.

This explains two linked observational effects near the transition region:

1. The one-component mean velocity can be driven toward the midpoint between the two LOSVDs because their opposite first moments cancel.
2. At the same time, the one-component dispersion becomes large because the two LOSVD centers are separated.

Thus a region with *small apparent* V *but enhanced* σ can be exactly where the underlying velocity separation is most important.

8 The useful 50/50, equal-dispersion approximation

For target screening, we often want a simple and deliberately conservative approximation. Assume that near the 2σ peak

$$f_A = 0.5$$

and that the two disks have approximately the same intrinsic dispersion,

$$\sigma_A \simeq \sigma_B \simeq \sigma_0.$$

Then the full mixture equation becomes

$$\begin{aligned} \sigma_{\text{peak}}^2 &= 0.5\sigma_0^2 + 0.5\sigma_0^2 + 0.25\Delta V^2 \\ &= \sigma_0^2 + \frac{\Delta V^2}{4}. \end{aligned}$$

Therefore

$$\boxed{\sigma_{\text{peak}}^2 = \sigma_0^2 + \frac{\Delta V^2}{4}.}$$

Rearranging,

$$\begin{aligned}\sigma_{\text{peak}}^2 - \sigma_0^2 &= \frac{\Delta V^2}{4},\\ 4\left(\sigma_{\text{peak}}^2 - \sigma_0^2\right) &= \Delta V^2,\end{aligned}$$

and hence

$$\boxed{\Delta V = 2\sqrt{\sigma_{\text{peak}}^2 - \sigma_0^2}.}$$

For candidate selection, we identify σ_0 with a plausible off-peak intrinsic disk dispersion and write

$$\boxed{\Delta V_{\text{cons}} = 2\sqrt{\sigma_{\text{peak}}^2 - \sigma_{\text{base}}^2}.}$$

The subscript "cons" emphasizes that this is a conservative separation estimate *within the adopted two-disk model*, not a model-independent measurement.

9 Why use the off-peak dispersion as σ_{base}?

The goal is to estimate how broad each disk would be before the extra broadening caused by counter-rotation becomes important. A practical proxy is the reliable stellar dispersion measured along or near the kinematic major axis outside the 2σ enhancement, in regions where one component is expected to dominate more strongly.

The logic is:

$$\underbrace{\sigma_{\text{peak}}^2}_{\text{observed variance at the peak}} - \underbrace{\sigma_{\text{base}}^2}_{\text{ordinary disk variance}} \approx \underbrace{\frac{\Delta V^2}{4}}_{\text{extra variance from separation, for 50/50 light}}.$$

This is preferable to using the visible amplitude of the single-component MaNGA velocity map as though it directly measured V_A or V_B. Around the transition region, the single-component velocity is itself a blended first moment and can be strongly suppressed.

9.1 Why choosing a high plausible σ_{base} is conservative

For fixed σ_{peak},

$$\Delta V_{\text{cons}} = 2\sqrt{\sigma_{\text{peak}}^2 - \sigma_{\text{base}}^2}.$$

A larger adopted σ_{base} leaves less excess variance to explain with counter-rotation, so it produces a smaller inferred separation. This is why it is sensible to use the *larger reliable* off-peak dispersion rather than suspiciously low values from very large/noisy outer Voronoi bins.

Figure 3 shows this for a representative $\sigma_{\text{peak}} = 101\ \text{km}\,\text{s}^{-1}$.

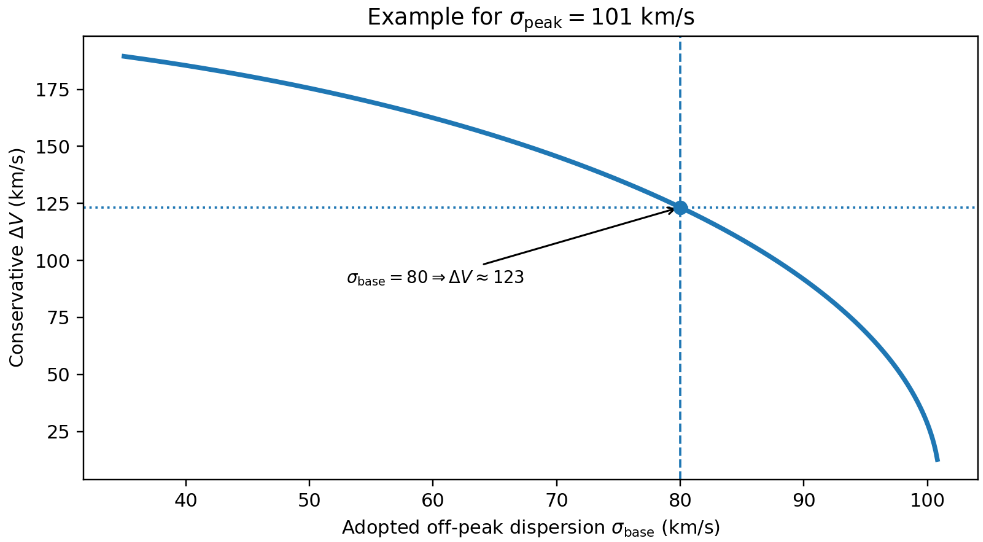

Figure 3: For a fixed σ_{peak}, adopting a higher plausible σ_{base} gives a smaller and therefore more conservative inferred velocity separation.

10 Worked example: 10838-1902

For the candidate 10838-1902, a representative reading of the maps was

$$\sigma_{\text{peak}} \simeq 101 \text{ km s}^{-1}, \qquad \sigma_{\text{base}} \simeq 80 \text{ km s}^{-1}.$$

Using the high plausible off-peak value intentionally makes the estimate conservative.

The excess variance is

$$\begin{aligned} \sigma_{\text{peak}}^2 - \sigma_{\text{base}}^2 &= 101^2 - 80^2 \\ &= 10201 - 6400 \\ &= 3801 \ (\text{km s}^{-1})^2. \end{aligned}$$

For equal light,

$$\frac{\Delta V^2}{4} = 3801,$$

so

$$\Delta V = 2\sqrt{3801} = 123.3 \text{ km s}^{-1}.$$

Thus the conservative model is approximately

$$\boxed{\Delta V_{\text{cons}} \simeq 123 \text{ km s}^{-1}.}$$

If the systemic-rest-frame single-component velocity near the transition is approximately zero, the clean symmetric realization is

$$\boxed{V_A \simeq +61.5 \text{ km s}^{-1}, \qquad V_B \simeq -61.5 \text{ km s}^{-1}.}$$

and for the injection-recovery experiment we would use

$$\boxed{\sigma_A = \sigma_B \simeq 80 \text{ km s}^{-1}.}$$

If instead the local blended mean is reliably $V_{\mathrm{mean}} \neq 0$, center the pair around it:

$$\boxed{V_A = V_{\mathrm{mean}} + \frac{\Delta V}{2}, \qquad V_B = V_{\mathrm{mean}} - \frac{\Delta V}{2}.}$$

The important point is that V_{mean} is the local midpoint of the mixture; it is not itself one of the two component velocities.

11 Why this can identify more viable CRD targets

A screening procedure based only on the apparent single-component MaNGA velocity amplitudes can reject good targets for exactly the wrong reason. Near a roughly equal-light transition, the two disks cancel in the first moment:

$$\bar{V} = f_A V_A + (1 - f_A) V_B.$$

Consequently, a weak apparent velocity pattern does not necessarily imply a small underlying ΔV.

The second moment contains complementary information. If the off-peak disk is relatively narrow but the transition region shows a robust, spatially coherent 2σ enhancement, then the excess variance can imply a much larger separation than the single-Gaussian V map suggests.

As a simple illustration, suppose a candidate has

$$\sigma_{\mathrm{peak}} = 100, \qquad \sigma_{\mathrm{base}} = 60 \ \mathrm{km\,s^{-1}}.$$

Then

$$\Delta V_{\mathrm{cons}} = 2\sqrt{100^2 - 60^2} = 160 \ \mathrm{km\,s^{-1}}.$$

A galaxy whose apparent blended velocity map looks modest could therefore still be an excellent high-resolution decomposition target.

This is particularly relevant for RH3 because the final question is not whether MaNGA itself cleanly resolves both LOSVD peaks. The question is whether a physically plausible separation implied by the MaNGA first and second moments will be recoverable with the much narrower KCRM instrumental LSF and the planned high-S/N observations.

12 Recommended target-selection workflow

For each candidate CRD, the following procedure provides a consistent screening estimate.

1. **Place the stellar velocity field in a sensible systemic frame.** Use the fitted stellar V_{sys} or equivalent kinematic zero point consistently.

2. **Identify the 2σ region.** Determine the approximate radial location and spatial extent of the off-center dispersion enhancement.

3. **Measure σ_{peak}.** Use representative, reliable bins in the coherent peak rather than a single isolated noisy maximum.

4. **Measure σ_{base}.** Inspect reliable off-peak bins along or near PA_{kin}, preferably where the 2σ enhancement has subsided and one component plausibly dominates. Avoid letting very large/noisy outer bins set the baseline solely because they have the lowest numerical σ.

5. **Adopt the larger plausible baseline for a conservative test.** This assigns as much of the peak width as reasonably possible to intrinsic disk dispersion and therefore minimizes the inferred ΔV.

6. **Compute the conservative separation.**

$$\boxed{\Delta V_{\rm cons} = 2\sqrt{\sigma_{\rm peak}^2 - \sigma_{\rm base}^2}.}$$

If $\sigma_{\rm peak} \le \sigma_{\rm base}$, this particular diagnostic provides no positive separation constraint; do not force the formula.

7. **Choose the velocity midpoint.** Read the local systemic-corrected single-component velocity around the peak/transition, $V_{\rm mean}$. If it is consistent with zero, use a symmetric pair. Otherwise use

$$V_A = V_{\rm mean} + \Delta V/2, \qquad V_B = V_{\rm mean} - \Delta V/2.$$

8. **Set the injected intrinsic dispersions.** For the conservative baseline experiment, use

$$\sigma_A = \sigma_B = \sigma_{\rm base}.$$

9. **Set $f_A = 0.5$ for the peak-region screening case.** This maximizes the separation-induced variance for a given ΔV, thereby minimizing the ΔV required to explain the peak. It also approximates the region where a classic 2σ signature is expected to be strongest.

10. **Run RH3 injection/recovery at the planned S/N.** Treat this as the conservative feasibility case. If it succeeds, plausible colder disks or unequal light fractions generally require a larger separation to produce the same $\sigma_{\rm peak}$, which tends to make the kinematic decomposition easier.

13 Important assumptions and caveats

The mixture-variance equation itself is exact for the variance of a two-component probability mixture with the stated component means, variances, and weights. The *target-selection inference* from MaNGA maps is approximate and depends on additional astrophysical and measurement assumptions.

13.1 The observed MaNGA σ is not guaranteed to equal the exact mixture moment

A single-Gaussian pPXF fit to a non-Gaussian LOSVD is an approximation. Its returned V and σ should be close to the dominant first-two-moment description, but template weighting, finite spectral resolution, masking, and the detailed LOSVD shape can shift the fitted values.

13.2 Use the appropriate instrumental-resolution correction

The $\sigma_{\rm peak}$ and $\sigma_{\rm base}$ used in this argument should represent astrophysical stellar broadening under the same convention. Do not compare an uncorrected observed width in one region to an LSF-corrected intrinsic width in another.

13.3 $\sigma_{\rm base}$ may vary with radius for real dynamical reasons

A bulge, central heating, asymmetric drift, or genuine radial changes in disk dispersion can change σ independently of counter-rotation. The off-peak baseline should therefore be chosen from regions that are dynamically relevant to the peak, not from an arbitrary distant bin.

13.4 Beam smearing and the PSF can also raise σ

Unresolved velocity gradients broaden the spectrum. A strong $\sigma_{\rm peak}$ caused partly by beam smearing would make the simple two-disk inference overestimate the counter-rotation contribution. Spatially paired off-center peaks and coherent 2σ morphology are more convincing than a single central dispersion maximum.

13.5 The two disks need not have equal intrinsic dispersions

The general equation remains

$$\sigma_{\mathrm{mix}}^2 = f_A\sigma_A^2 + (1 - f_A)\sigma_B^2 + f_A(1 - f_A)\Delta V^2.$$

The equal-σ approximation is a screening simplification. Once real RH3 data exist, σ_A and σ_B should be fit separately.

13.6 $f_A = 0.5$ is a conservative mathematical choice, not a measured fact

For fixed intrinsic dispersions and observed σ_{peak}, the required separation is

$$\Delta V = \sqrt{\frac{\sigma_{\mathrm{peak}}^2 - [f_A\sigma_A^2 + (1 - f_A)\sigma_B^2]}{f_A(1 - f_A)}}.$$

Because $f_A(1 - f_A)$ is largest at 0.5, the equal-light assumption minimizes the separation required by the model. The actual local light ratio may differ.

14 What this calculation does and does not establish

What it does: It converts a coherent excess in the stellar second moment into a physically motivated, conservative characteristic LOSVD separation *under the two-counterrotating-disk interpretation.* This is useful for injection/recovery simulations and target ranking.

What it does not do: It does not uniquely decompose the MaNGA spectrum, prove that every bit of excess σ comes from counter-rotation, or provide exact V_A, V_B, σ_A, and σ_B. Those are the quantities the higher-resolution RH3 observations and full likelihood-cube analysis are intended to determine.

15 Compact reference

For future candidate screening, the core equations are:

Combined mean velocity

$$\boxed{\bar{V} = f_A V_A + (1 - f_A)V_B.}$$

Combined two-component variance

$$\boxed{\sigma_{\mathrm{mix}}^2 = f_A\sigma_A^2 + (1 - f_A)\sigma_B^2 + f_A(1 - f_A)\Delta V^2.}$$

Equal-light, equal-dispersion peak approximation

$$\boxed{\sigma_{\mathrm{peak}}^2 = \sigma_{\mathrm{base}}^2 + \frac{\Delta V^2}{4}.}$$

Conservative separation used for screening

$$\boxed{\Delta V_{\mathrm{cons}} = 2\sqrt{\sigma_{\mathrm{peak}}^2 - \sigma_{\mathrm{base}}^2}.}$$

Velocity pair centered on the local blended mean

$$\boxed{V_A = V_{\mathrm{mean}} + \frac{\Delta V}{2}, \qquad V_B = V_{\mathrm{mean}} - \frac{\Delta V}{2}.}$$

Conservative injection setup near a 2σ peak

$$\boxed{f_A = 0.5, \qquad \sigma_A = \sigma_B = \sigma_{\mathrm{base}}.}$$

The central idea to remember: *a blended LOSVD can be broad either because each component is intrinsically broad, because the component means are far apart, or both. The 2σ method uses reliable off-peak dispersions to estimate how much of the peak width is left for the separation term.*